

The Exponential Growth Constant of OEIS A158415

Existence, an effective characterization,
and certified numerical bounds

Research article

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Abstract

Let $a(n)$ count the distinct real values represented by expressions with exactly n symbols, using the constant 1, principal square root, and binary addition. Numerical equality, rather than equality of expression trees, is essential. We prove that a computable constant $\gamma > 1$ exists such that $\log a(n) = n \log \gamma + o(n)$, and obtain the certified bounds $1.86204 < \gamma < 1.9208$. The main algebraic step is a uniform injectivity theorem: any finite alphabet of representable values can be used as digits in nested radicals after a suitable fixed seed is chosen. Its proof uses an odd valuation at a prime above 2 and irreducible polynomials of controlled degree. A complementary decomposition of minimum-size expressions yields convergent finite-data upper bounds. Together these arguments also prove the exact critical identity $\sum_{n \geq 1} a(n) \gamma^{-n} = \gamma^2$; in particular, $a(n) = o(\gamma^n)$. The numerical lower bound is backed by an independently replayable certificate for 1,526,536 pairwise distinct values, using only integer arithmetic. A separate combinatorial majorant certifies the upper bound and, after local simplifications, proves the first 21 listed sequence terms exactly. The value of γ beyond the stated interval, a ratio limit, and a possible $n^{-3/2}$ prefactor are not determined here.

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1 The sequence and the question

An expression is built by the grammar

$$E ::= 1 \mid \sqrt{E} \mid (E + E).$$

Its size is the number of occurrences of 1, square-root operators, and addition operators. Parentheses are not counted. Thus

$$|1| = 1, \quad |\sqrt{E}| = |E| + 1, \quad |E + F| = |E| + |F| + 1.$$

Every square root denotes the nonnegative real square root. All values produced by this grammar are at least 1. The sequence in the question is OEIS A158415 [1], introduced by Vladimir Reshetnikov in 2009. We study its defining counting problem, not just an extrapolation of its recorded initial values.

Write $\mathcal{S}$ for the set of all represented values and define

$$c(x) = \min\{|E| : E \text{ represents } x\}, \quad x \in \mathcal{S}.$$

Replacing a leaf 1 by $\sqrt{1}$ increases the size by one without changing the value. Consequently, the exact-size and cumulative interpretations agree:

$$a(n) = \#\{x \in \mathcal{S} : c(x) \leq n\}. \quad (1.1)$$

Set $a(0) = 0$ and

$$b(n) = a(n) - a(n-1) = \#\{x \in \mathcal{S} : c(x) = n\}. \quad (1.2)$$

The first few values are easy to check. At size at most four the values are 1, 2, and $\sqrt{2}$, so $a(4) = 3$. The main difficulty is that different syntactic expressions can have the same value. Besides associativity and commutativity, one has identities such as

$$\sqrt{1+1+1+1} = 1+1, \quad \sqrt{3+2\sqrt{2}} = 1+\sqrt{2}.$$

Here integer coefficients and integer constants are abbreviations for sums of copies of 1; they are not additional symbols in the grammar.

The original sequence discussion explicitly raised the issue of proving numerical equality and questioned the fixed tolerance used in a PARI calculation [2]. That distinction remains important here: our numerical certificates never infer equality merely from overlapping approximations.

2 Main result and its scope

Theorem 2.1 (Growth constant and effective characterization). *There is a computable real number γ such that*

$$\lim_{n \rightarrow \infty} a(n)^{1/n} = \gamma, \quad a(n) = \gamma^{n+o(n)}. \quad (2.1)$$

It satisfies

$$1.86204 < \gamma < 1.9208. \quad (2.2)$$

For each $m \geq 1$, let $r_m \in (0, 1)$ be the unique solution of

$$r_m \left(1 + \sum_{j=1}^m b(j) r_m^{j+1} \right) = 1, \quad \lambda_m = r_m^{-1}. \quad (2.3)$$

Then λ_m is nondecreasing and

$$\gamma = \sup_{m \geq 1} \lambda_m = \lim_{m \rightarrow \infty} \lambda_m. \quad (2.4)$$

There are also explicitly defined algebraic upper bounds $\mu_h \geq \gamma$, computed from $b(1), \dots, b(3h-1)$, with $\mu_h \rightarrow \gamma$. Finally,

$$\boxed{\sum_{n \geq 1} a(n) \gamma^{-n} = \gamma^2}, \quad a(n) = o(\gamma^n). \quad (2.5)$$

The finite lower-bound equation can equivalently be written as the polynomial identity

$$\lambda_m^{m+2} = \lambda_m^{m+1} + \sum_{j=1}^m b(j) \lambda_m^{m-j}. \quad (2.6)$$

Thus the theorem gives more than a statement that the sequence lies between two exponential functions. It proves a single exponential rate, supplies convergent finite-data characterizations from both sides, and identifies an exact normalization of the generating function at its radius of convergence.

It does *not* give a closed form for γ , prove a limit for $a(n+1)/a(n)$, or establish an equivalent of the form $C\gamma^n n^{-3/2}$. These stronger assertions must not be inferred from the root limit. In fact, (2.5) excludes a positive constant prefactor $C\gamma^n$. More generally, if $a(n) \sim C\gamma^n n^{-\theta}$ with $C > 0$, then necessarily $\theta > 1$.

3 Elementary structural facts

3.1 Size, magnitude, and algebraic integrality

Lemma 3.1. *Every represented value is an algebraic integer. If an expression has L leaves and U square-root nodes, then its size is $2L - 1 + U$ and every conjugate of its value has absolute value at most L .*

Proof. The number of binary nodes is $L - 1$, giving the size formula. Algebraic integers are closed under addition, and a square root of an algebraic integer is integral over the ring of algebraic integers and hence over $\mathbb{Z}$. For the conjugate bound, put all subexpression values in a finite number field and apply any embedding into $\mathbb{C}$. Addition obeys the triangle inequality. For a radical node, $|\sigma(\sqrt{x})| = \sqrt{|\sigma(x)|} \leq \sqrt{L} \leq L$. The assertion follows by induction from the leaf value 1. $\square$

In particular, $c(k) = 2k - 1$ for each positive integer k : at least k leaves are required to attain magnitude k , and their sum realizes this size. There are therefore new values at arbitrarily large sizes.

A prefix encoding of an expression is a word over three symbols. This gives the coarse but sufficient initial estimate

$$1 \leq a(n) \leq 3^n. \quad (3.1)$$

3.2 Multiplication is available at finite cost

Multiplication is not an allowed symbol. Nevertheless its values are available. This observation is needed for the number-field argument, not as a claim that multiplication is inexpensive.

Lemma 3.2. *If $x, y \in \mathcal{S}$, then $xy \in \mathcal{S}$.*

Proof. Induct on the sum of the sizes of chosen expressions for x and y . A factor 1 is harmless. If either expression is a sum, distribute the product and use the induction hypothesis. Otherwise both nontrivial factors are radicals, and

$$(\sqrt{u})(\sqrt{v}) = \sqrt{uv}.$$

The product uv involves smaller expressions, so it too is representable. $\square$

It follows that every element of a ring $\mathbb{Z}[d_1, \dots, d_s]$ with $d_i \in \mathcal{S}$ can be written as $A - B$, where $A, B \in \mathcal{S} \cup \{0\}$. If desired, adding 1 to both parts makes A and B genuine elements of $\mathcal{S}$.

4 Injective coding by nested radicals

This section is the central algebraic argument. It overcomes the fact that ordinary addition is not an injective pairing operation.

We use standard facts about orders in number fields and discrete valuations: if K is a number field and $\mathfrak{p}$ lies above 2, then the normalized valuation $v = v_{\mathfrak{p}}$ takes integer values on $K^\times$, its residue field is finite of characteristic 2, and it extends to algebraic extensions. An order in K has finite index in $\mathcal{O}_K$. These facts, and the corresponding Newton-polygon viewpoint, are treated in [3, Chapters 2, 3, and 7]. The particular irreducibility argument below is proved directly.

4.1 A representable element with odd valuation

Lemma 4.1. *Let $D \subset \mathcal{S}$ be finite and $K = \mathbb{Q}(D)$. There are a prime $\mathfrak{p}$ of K above 2 and an element $\pi \in \mathcal{S} \cap K$ such that $h = v_{\mathfrak{p}}(\pi)$ is a positive odd integer.*

Proof. Choose any prime $\mathfrak{p}$ above 2, and write $e = v_{\mathfrak{p}}(2)$. If e is odd, take $\pi = 2$.

Suppose e is even. The ring $R = \mathbb{Z}[D]$ is an order in K : it is generated by algebraic integers and spans K over $\mathbb{Q}$. Choose a positive integer I with $I\mathcal{O}_K \subset R$, and choose $u \in \mathcal{O}_K$ with $v(u) = 1$. Then

$$t = Iu \in R, \quad v(t) = e v_2(I) + 1 =: h,$$

so h is odd. By Lemma 3.2, write $t = A - B$ with $A, B \in \mathcal{S}$. For a sufficiently large integer M , $v(2^M B) > h$. The positive, representable number

$$\pi = A + (2^M - 1)B = t + 2^M B$$

then has valuation h . Integer multiples here are realized by repeated addition, so the construction stays inside the original language. $\square$

The element π need not be a uniformizer. Allowing an arbitrary positive *odd* valuation is what makes the construction work without requiring the given order to be integrally closed at 2.

4.2 A local irreducibility lemma

Lemma 4.2. *Let K be as above, let $v(\pi) = h > 0$ be odd, and choose a power of two $N = 2^t > h$. Suppose $Q(X) \in \mathcal{O}_K[X]$ is monic of degree $d = 2^r$ and its reduction modulo $\mathfrak{p}$ has the form $X^d + c$. Then*

$$Q(X)^N - \pi \tag{4.1}$$

is irreducible over K .

Proof. The residue field is finite, so the map $x \mapsto x^d$ is bijective. Choose a residue-field root of $X^d + c$ and lift it to $a \in \mathcal{O}_K$. For $q(X) = Q(X + a)$, reduction modulo $\mathfrak{p}$ is exactly X^d . Thus q is monic and every nonleading coefficient has valuation at least one.

Extend v to an algebraic closure and let α be any root of $q(X)^N - \pi$. Then $v(q(\alpha)) = h/N \in (0, 1)$. If $v(\alpha) < 0$, the leading term of $q(\alpha)$ has uniquely smallest valuation, giving a negative valuation, a contradiction. If $v(\alpha) \geq 1/d$, every term of $q(\alpha)$ has valuation at least one, also a contradiction. For $0 \leq v(\alpha) < 1/d$, the leading term again has uniquely smallest valuation, and therefore

$$v(\alpha) = \frac{h}{Nd}. \tag{4.2}$$

In particular $v(\alpha) > 0$.

If the polynomial had a monic factor of degree k over K , the constant coefficient of that factor would be, up to sign, a product of k roots. Its valuation would be $kh/(Nd)$, which must be an integer. Since h is odd and Nd is a power of two, this implies $Nd \mid k$. No nonconstant proper factor is possible. Translating back proves the claim. $\square$

4.3 One seed works for the whole finite alphabet

Theorem 4.3 (Uniform injectivity of radical spines). *For every finite set $D \subset \mathcal{S}$ there is a representable positive seed s such that all the numbers*

$$d_0 + \sqrt{d_1 + \sqrt{\cdots + \sqrt{d_r + s}}}, \quad r \geq 0, \quad d_i \in D \cup \{0\}, \tag{4.3}$$

are distinct as the complete digit string $(d_0, \dots, d_r)$ varies. For $r = 0$ the expression means $d_0 + s$. A zero digit means that the corresponding addition is omitted.

Proof. Take $K = \mathbb{Q}(D)$, choose π by Lemma 4.1, and put $s = \pi^{1/N}$, where $N = 2^t > v(\pi)$. This is a fixed representable seed, obtained by t successive square roots.

For the string in (4.3), define

$$Q_0(X) = X - d_0, \quad Q_j(X) = Q_{j-1}(X)^2 - d_j \quad (1 \leq j \leq r).$$

The resulting value x satisfies $Q_r(x) = s$, so it is a root of

$$P_d(X) = Q_r(X)^N - \pi.$$

Modulo $\mathfrak{p}$, repeated squaring gives a polynomial of the form $X^{2^r} + c$. Lemma 4.2 shows that P_d is monic irreducible of degree $N2^r$.

Equal values must have equal monic minimal polynomials over K . Their degrees first force the same r . Equality of the minimal polynomials then gives $Q_r^N = \tilde{Q}_r^N$, hence $Q_r = \tilde{Q}_r$, since the polynomials are monic. The decomposition $Q_j = Q_{j-1}^2 - d_j$ is unique: if $A^2 - u = B^2 - v$ with A, B monic of the same positive degree, then $(A - B)(A + B) = u - v$ is constant. Unless $A = B$, its left side has positive degree. Thus $A = B$ and $u = v$. Peeling off the successive squares recovers all digits, including d_0 . $\square$

Remark 4.4. The seed can have a very large expression size. This is harmless for an asymptotic lower bound with a fixed finite alphabet, but it prevents the proof from being interpreted as a practical short encoding of arbitrary finite sets. The order of limits is important: first fix the alphabet and let the output size tend to infinity; only then enlarge the alphabet.

5 Existence of the exponential rate

5.1 Finite alphabets give genuine exponential lower bounds

Assign one expression of size $\ell(d)$ to each digit in a finite alphabet D . Adding a nonzero digit costs $\ell(d) + 1$ symbols, whereas the zero digit costs nothing. Define

$$D_\ell(z) = 1 + \sum_{d \in D} z^{\ell(d)+1}. \quad (5.1)$$

If the seed of Theorem 4.3 has an expression of size L , then all its spines have the exact generating function

$$z^L \sum_{r \geq 0} z^r D_\ell(z)^{r+1} = \frac{z^L D_\ell(z)}{1 - z D_\ell(z)}. \quad (5.2)$$

This counts *distinct values*, not merely formal strings, by the injectivity theorem. Summing over expression sizes at most n introduces an additional factor $(1 - z)^{-1}$.

Let $\rho \in (0, 1)$ solve $\rho D_\ell(\rho) = 1$. The zero of $1 - z D_\ell(z)$ at ρ is simple, and it is the only zero on its circle of convergence. Indeed equality in the triangle inequality would require the degree-one term of $z D_\ell(z)$ to have positive real phase, forcing $z = \rho$. Partial fractions, or the usual simple-pole coefficient calculation, therefore gives

$$a(n) \geq C_D \rho^{-n} \quad (n \text{ sufficiently large}) \quad (5.3)$$

for a constant $C_D > 0$. The standard rational-function framework is explained in [4, Chapters IV–V]; here the needed pole properties are explicit.

Taking all values with $c(d) \leq m$, and assigning their minimum sizes, gives the lower bases λ_m in (2.3). In particular,

$$\liminf_{n \rightarrow \infty} a(n)^{1/n} \geq \sup_m \lambda_m. \quad (5.4)$$

5.2 The generating-function argument that removes oscillation

Define

$$A(z) = \sum_{n \geq 1} a(n) z^n, \quad B(z) = \sum_{n \geq 1} b(n) z^{n+1} = z(1 - z)A(z). \quad (5.5)$$

Put $g = \sup_m \lambda_m$. It is finite by (3.1) and (5.4), and $g > 1$ already follows from the alphabet $\{1\}$. For every $0 < z \leq 1/g$ and every m ,

$$z \left(1 + \sum_{j=1}^m b(j) z^{j+1} \right) \leq 1.$$

Monotone convergence in m shows that $B(z)$ is finite and

$$z(1 + B(z)) \leq 1 \quad (0 < z \leq 1/g). \quad (5.6)$$

Thus $A(z)$ is finite there as well. By the elementary radius formula for a power series with nonnegative coefficients,

$$\limsup_{n \rightarrow \infty} a(n)^{1/n} \leq g.$$

Together with (5.4), this proves the root limit, with $\gamma = g$. Writing $R = 1/\gamma$, we have established

$$B(R) < \infty, \quad R(1 + B(R)) \leq 1, \quad A(R) \leq \gamma^2. \quad (5.7)$$

The inequality will become an equality in Section 6. Already it implies $a(n)\gamma^{-n} \rightarrow 0$.

5.3 An elementary explicit subfamily

For integer digits, a single seed works for the entire infinite alphabet: take $K = \mathbb{Q}$, $\pi = 2$, $N = 2$, and $s = \sqrt{2}$, of size four. The proof of Theorem 4.3 applies uniformly to all nonnegative integer digits. A digit k costs $2k$ symbols when added to a spine, so $D(z) = (1 - z^2)^{-1}$. Formula (5.2) becomes

$$\frac{z^4}{1 - z - z^2}.$$

With $F_1 = F_2 = 1$, cumulative counting gives the entirely explicit estimate

$$a(n) \geq F_{n-1} - 1 \quad (n \geq 4). \quad (5.8)$$

In particular, $\gamma \geq (1 + \sqrt{5})/2$. The numerical certificate below improves this substantially.

6 Finite-data upper bounds and computability

The matching upper mechanism uses addition's associativity. It is here that counting numerical values, with minimum expression sizes, supplies useful structure rather than just an obstacle.

6.1 Additive weights and minimum subexpressions

Give an expression the weight $w(E) = |E| + 1$, and a value the minimum weight $w(x) = c(x) + 1$. Then

$$w(E + F) = w(E) + w(F), \quad w(\sqrt{E}) = w(E) + 1.$$

Flatten every maximal addition into a multiset of atoms, where an atom is 1 or a square-root expression. If the entire expression has minimum size, every subexpression is minimum. More strongly, the sum of *any nonempty submultiset* of a flattened addition is minimum for its value.

Otherwise one could regroup that submultiset, replace it by a smaller expression, and shorten the whole expression. Regrouping does not change the number of addition symbols.

For an integer $h \geq 2$, define the finite polynomials

$$\mathcal{D}_h(z) = 1 + \sum_{n+1 \leq h} b(n)z^{n+1}, \quad (6.1)$$

$$\mathcal{T}_h(z) = \sum_{h < n+1 \leq 3h} b(n)z^{n+1}. \quad (6.2)$$

Lemma 6.1 (Large-expression decomposition). *Every minimum-size expression whose weight is greater than $3h$ can be decomposed, without changing its weight, in one of two ways:*

- (i) as $d + \sqrt{E}$, where $d = 0$ or $w(d) \leq h$, and $w(E) > h$;
- (ii) as $E + F$, where both $w(E)$ and $w(F)$ exceed h .

All the nonzero components may be represented with their minimum weights.

Proof. A radical at the root gives (i) with $d = 0$. Otherwise flatten the top addition. If its largest atom has weight greater than h , consider the total weight of the remaining atoms. If that remainder exceeds h , split it off to obtain (ii). If it is at most h , the large atom cannot be 1 and is therefore a radical, giving (i).

If every atom has weight at most h , collect atoms until their total weight first exceeds h . This total is at most $2h$, and the remaining atoms have weight greater than h because the whole weight exceeds $3h$. This gives (ii). Minimum-weight status follows from the preceding submultiset argument. In case (i), the radical child has weight greater than $2h - 1$, hence greater than h . $\square$

Stop the decomposition when a component has weight in $(h, 3h]$. The resulting codes are counted, with possible overcounting, by the nonnegative formal power series H_h satisfying

$$H_h = \mathcal{T}_h + z\mathcal{D}_h H_h + H_h^2, \quad H_h(0) = 0. \quad (6.3)$$

The last term uses ordered children; allowing both orders only enlarges the upper count. Each value of weight greater than h has at least one code of its minimum weight, and a code determines a value. Therefore

$$\sum_{n+1 > h} b(n)z^{n+1} \preceq H_h(z), \quad (6.4)$$

where $\preceq$ denotes coefficientwise domination.

6.2 Explicit algebraic upper bounds

Let s_h be the first positive solution, before $z\mathcal{D}_h(z) = 1$, of

$$(1 - z\mathcal{D}_h(z))^2 = 4\mathcal{T}_h(z), \quad \mu_h = s_h^{-1}. \quad (6.5)$$

Such a solution is unique: before $z\mathcal{D}_h(z) = 1$ the left side decreases strictly and the right side increases. Also $\mathcal{T}_h$ is nonzero, since positive integers have minimum weights $2, 4, 6, \dots$. The smaller solution of (6.3) is

$$H_h(z) = \frac{1 - z\mathcal{D}_h(z) - \sqrt{(1 - z\mathcal{D}_h(z))^2 - 4\mathcal{T}_h(z)}}{2}. \quad (6.6)$$

It is finite at s_h and has nonnegative coefficients. For example, bounded iteration of (6.3) proves convergence for positive $z \leq s_h$ without any assumptions about other complex singularities. Equation (6.4) gives

$$\gamma \leq \mu_h. \quad (6.7)$$

Proposition 6.2. *The upper bounds satisfy $\mu_h \rightarrow \gamma$. Consequently*

$$\gamma = \sup_m \lambda_m = \inf_{h \geq 2} \mu_h.$$

The running minima of μ_h form a nonincreasing sequence of upper bounds.

Proof. Fix $0 < z < R = 1/\gamma$. By (5.7) and strict monotonicity, $z(1 + B(z)) < 1$. As $h \rightarrow \infty$,

$$\mathcal{D}_h(z) \rightarrow 1 + B(z), \quad \mathcal{T}_h(z) \rightarrow 0.$$

Thus $(1 - z\mathcal{D}_h(z))^2 > 4\mathcal{T}_h(z)$ for all sufficiently large h , and $s_h > z$. Hence $\limsup_h \mu_h \leq 1/z$. Letting $z \uparrow R$ and using (6.7) proves the result. $\square$

6.3 The exact critical identity

Proposition 6.3. *At $R = 1/\gamma$,*

$$R(1 + B(R)) = 1. \quad (6.8)$$

Equivalently, $A(R) = R^{-2}$.

Proof. We already know the left side is at most one and $B(R)$ is finite. Suppose it were strictly less than one. Then, as $h \rightarrow \infty$, $\mathcal{T}_h(R) \rightarrow 0$ and $1 - R\mathcal{D}_h(R)$ stays bounded away from zero. For some finite h we would have

$$1 - R\mathcal{D}_h(R) > 0, \quad (1 - R\mathcal{D}_h(R))^2 > 4\mathcal{T}_h(R).$$

Both inequalities are strict polynomial inequalities. They would remain true at some $z > R$, implying convergence of $H_h(z)$ and hence of $B(z)$. This contradicts the definition of R as the radius of convergence. Therefore (6.8) holds. Substitution of $B(R) = R(1 - R)A(R)$ gives $A(R) = R^{-2}$. $\square$

6.4 Why this is an effective characterization

Every finite set of expression values is exactly computable using algebraic number arithmetic; a completely explicit separation bound is proved in Section 10. Thus all $b(1), \dots, b(M)$ are computable in finite time. Positive roots of the finite polynomials (2.3) and (6.5) can be bracketed by rational arithmetic.

To approximate γ within a prescribed positive error ε , compute increasing finite sets of exact terms and the associated lower bounds λ_m and running upper bounds $\min_{2 \leq h \leq H} \mu_h$. Stop when the two rational brackets differ by less than ε . Proposition 6.2 proves that this algorithm terminates. No unproved convergence rate or numerical extrapolation is needed. This is a computability statement, not a polynomial-time claim.

7 A certified numerical lower bound

7.1 What the certificate proves

The archive contains a directed acyclic graph of 1,526,536 expressions, each with an explicitly recorded symbol cost at most 28. Reusing a stored node is only a compression technique: the cost calculation charges a child again every time it is used, so this is *formula size*, not circuit size.

For each node the independent verifier computes an interval $[\ell/2^{256}, u/2^{256}]$ using integer arithmetic. For addition it adds endpoints. For a square root it uses

$$\ell' = \left\lfloor \sqrt{\ell 2^{256}} \right\rfloor, \quad u' = \left\lceil \sqrt{u 2^{256}} \right\rceil. \quad (7.1)$$

An integer square-root routine computes these bounds exactly. After sorting all intervals, the verifier checks that every consecutive pair is strictly disjoint. This proves pairwise distinctness of all stored values. The smallest certified gap in this sample is approximately $5.46212307648 \times 10^{-13}$; the exact rational gap is recorded in the verification output.

The generator discards an interval whenever it overlaps one already retained. Such a discard is conservative: it is *not* an assertion that the two values are equal. Therefore the output is unconditionally a lower-bound sample, even if a very close distinct value were discarded.

Let q_n be the number of sample nodes assigned cost n . They need not be assumed to have minimum cost. Theorem 4.3 can use these assigned costs directly, so the positive root of

$$z \left(1 + \sum_{n=1}^{28} q_n z^{n+1} \right) = 1 \quad (7.2)$$

gives a valid lower bound on γ . In particular, no completeness claim about the enumeration is required.

7.2 The finite-root calculation

The cumulative sample counts agree with the 28 values currently listed in the OEIS entry [1]. For the lower-bound proof, however, the stored expressions and interval verification replace reliance on those listed numbers. The reciprocal root of (7.2) is bracketed by exact rational bisection and begins

$$1.86204402268771032792879 \dots \quad (7.3)$$

At the exact rational number $z = 25000/46551 = 1/1.86204$, the left side of (7.2), minus one, is strictly positive; its value is approximately $1.09151356699874 \times 10^{-5}$. Since that left side is strictly increasing in z , this proves $\gamma > 1.86204$.

The constants in the accompanying eventual inequalities $a(n) \geq C_m \lambda_m^n$ are not computed: the general seed construction can be large. The certified conclusion concerns the limiting exponential rate.

Maximum assigned digit cost	Reciprocal root, displayed approximately
4	1.618033988749895
8	1.741298211719966
12	1.792996772315302
16	1.821892848098540
20	1.840189044447032
24	1.852813818187051
28	1.862044022687710

Table 1: Certified finite-alphabet lower bases. The archive records rational brackets, not just these decimal displays.

8 A certified upper bound below 1.921

8.1 An unordered-sum majorant

Consider a class of formal expressions in which maximal additions are flattened into multisets and a radical of a pure integer square is forbidden. Every minimum-size expression belongs to this class: replacing $\sqrt{q^2}$ by q is always shorter. Other numerical identities are ignored, so the class provides an upper bound.

Use weight $|E| + 1$, and let $U(z) = \sum_{j \geq 2} u_j z^j$ count these formal nonempty expressions. The atom 1 has weight two. A radical adds one to the weight of its child, except that the forbidden radicals contribute

$$P(z) = \sum_{q \geq 1} z^{2q^2+1}. \quad (8.1)$$

Consequently the atom series is $z^2 + zU(z) - P(z)$, and the multiset construction gives

$$1 + U(z) = \exp \left(\sum_{k \geq 1} \frac{z^{2k} + z^k U(z^k) - P(z^k)}{k} \right). \quad (8.2)$$

The formula can also be checked directly by multiplying $(1 - z^j)^{-p_j}$ over atom weights j , where p_j is the number of atoms of that weight; see [4, Chapter I] for the general multiset construction.

Minimum-size expressions inject into this formal class after one representative per value is selected. Therefore

$$a(n) \leq \sum_{j=2}^{n+1} u_j. \quad (8.3)$$

8.2 Exact coefficients by the Euler transform

Let $c_0 = 1$ and $c_n = u_n$ for $n \geq 1$. The atom counts are

$$p_n = \mathbf{1}_{n=2} + u_{n-1} - \#\{q \geq 1 : n = 2q^2 + 1\}.$$

Put $e_k = \sum_{d|k} d p_d$. Logarithmic differentiation of the product formula gives the integer recurrence

$$n c_n = \sum_{k=1}^n e_k c_{n-k}. \quad (8.4)$$

There is no circular dependence: p_n uses only u_{n-1} . The certificate computes these coefficients through $n = 200$, checking nonnegativity and exact divisibility at every step.

8.3 A convergence criterion proved by radical height

For $r > 0$ with $r^2 < 1/3$, define

$$\mathcal{H}(r) = \sum_{k \geq 2} \frac{r^{2k} + r^k U(r^k) - P(r^k)}{k}, \quad (8.5)$$

$$\mathcal{F}(r) = \log r + 1 + r^2 - r - P(r) + \mathcal{H}(r). \quad (8.6)$$

All evaluations $U(r^k)$ in this definition converge: the number of formal objects of weight j is at most 3^{j-1} , by prefix encoding of an associated ordered expression. Hence $U(x)$ converges for $x < 1/3$.

Proposition 8.1. *If*

$$r^2 + r < 1, \quad r^2 < 1/3, \quad \mathcal{F}(r) < 0, \quad (8.7)$$

then $U(r)$ is finite and $\gamma < 1/r$.

Proof. Let U_j count objects of radical height at most j . At height zero they are positive integers, so $1 + U_0(r) = (1 - r^2)^{-1} < 1/r$. Every pure integer square is already present at height zero; thus removing P in each later atom construction is legitimate and leaves nonnegative coefficients. The classes increase with height, and their union is the whole formal class.

Writing $C_j = 1 + U_j$, the $k = 1$ term of the multiset recurrence gives

$$C_{j+1}(r) \leq \exp(rC_j(r) + r^2 - r - P(r) + \mathcal{H}(r)).$$

For $k \geq 2$, the bounded-height classes were bounded by the full class at r^k , where convergence is already known. If $C_j(r) \leq 1/r$, the exponent is at most $1 + r^2 - r - P(r) + \mathcal{H}(r) < -\log r$. Induction gives $C_j(r) < 1/r$ for every j , uniformly. Taking the increasing limit proves convergence at r .

The estimates are strict and persist at a slightly larger positive argument: the small-argument series in (8.5) are locally uniformly convergent while $r^2 < 1/3$. The same height argument thus proves convergence at some $r' > r$. Equation (8.3) yields $\gamma \leq 1/r' < 1/r$. $\square$

8.4 Explicit bounds for every omitted tail

For $0 < x < 1/3$, the prefix bound gives

$$U(x) \leq \sum_{j=2}^N u_j x^j + \frac{3^N x^{N+1}}{1 - 3x}. \quad (8.8)$$

Use the finite lower bound $P_Q(x) = \sum_{q=1}^Q x^{2q^2+1} \leq P(x)$ wherever $-P(x)$ occurs. For the terms with $k > K$ in (8.5), omit the negative terms and use $U(x) \leq 3x^2/(1 - 3x)$. Their total is at most

$$\frac{r^{2(K+1)}}{(K+1)(1-r^2)} + \frac{3r^{3(K+1)}}{(K+1)(1-r^3)(1-3r^{K+1})}. \quad (8.9)$$

Finally, for $t = (1 - r)/(1 + r) \in (0, 1)$,

$$\log r = -2 \sum_{j \geq 0} \frac{t^{2j+1}}{2j+1}.$$

After J terms the positive sum's omitted tail is at most

$$\frac{t^{2J+1}}{(2J+1)(1-t^2)}. \quad (8.10)$$

These formulas reduce the entire verification to finitely many rational operations and an explicitly bounded series for the logarithm.

8.5 The numerical certificate

The program `certify_upper_bound.py` evaluates an upper bound for $\mathcal{F}(r)$ at the exact rational argument

$$r = \frac{1250}{2401} = \frac{1}{1.9208}$$

using $N = 200$, $K = 40$, $Q = 8$, and $J = 100$. All arithmetic is enclosed by outward-rounded dyadic intervals with 1024 fractional bits. Every tail is bounded by the displayed formulas. The resulting upper bound is strictly negative and is approximately

$$-8.442263650713085 \times 10^{-6}. \quad (8.11)$$

The initial height margin $1/r - (1 - r^2)^{-1}$ is positive, approximately 0.5489798948. The exact signed integer numerator of the upper endpoint is stored in the archive. Thus Proposition 8.1, rather than an unverified floating-point root finder, proves

$$\boxed{\gamma < 1.9208.}$$

Together with Section 7 this completes the numerical part of Theorem 2.1.

9 Sharper finite-term bounds and exact initial terms

Two further local simplifications significantly tighten the formal upper counts. They are useful for certifying initial terms even though the simpler majorant of Section 8 suffices for the stated growth bound.

9.1 Square factors in additive multiplicities

If every multiplicity in a radical's flattened radicand is divisible by q^2 , $q \geq 2$, then its value is $\sqrt{q^2 t} = q\sqrt{t}$. If $w(t)$ denotes the syntactic weight of the divided multiset, the old and new weights are respectively

$$q^2 w(t) + 1, \quad q(w(t) + 1).$$

The old weight exceeds the new one by $(q - 1)(qw(t) - 1) > 0$. No minimum-size expression can contain such an atom.

The allowed radicands therefore have squarefree gcd of their multiplicities. If $V(z)$ counts the resulting formal expressions, the radicand series with this restriction is

$$\sum_{q \geq 1} \mu(q) V(z^{q^2}), \quad (9.1)$$

where μ is the ordinary Moebius function. Indeed a multiset with multiplicity gcd g is counted with coefficient $\sum_{q^2|g} \mu(q)$, which is one precisely when g is squarefree. The coefficients of (9.1) are consequently nonnegative despite its alternating presentation.

9.2 A family of denesting identities

Put $\alpha_k = 2^{1/2^k}$ for $k \geq 1$. Its displayed expression has size $k + 3$. For $m \geq 1$, forbid the atom obtained from the expanded square

$$\sqrt{m^2 + 2m\alpha_k + \alpha_k^2} = m + \alpha_k. \quad (9.2)$$

The right side is strictly shorter. The atom on the left has weight

$$2m^2 + 8m + 4 + (2m + 1)k. \quad (9.3)$$

For $k = 1$, combine $\alpha_1^2 = 2$ with the integer part; the multiplicity gcd is one or two. For $k \geq 2$, the distinct atom α_{k-1} has multiplicity one. Thus none of these exclusions overlaps the square-factor restriction. Their children are allowed dyadic-root atoms, and distinct pairs (m, k) give distinct formal radicands.

Define

$$E(z) = \sum_{m \geq 1} \sum_{k \geq 1} z^{2m^2 + 8m + 4 + (2m + 1)k}. \quad (9.4)$$

The improved atom series is

$$J(z) = z^2 + z \sum_{q \geq 1} \mu(q) V(z^{q^2}) - z^3 - E(z), \quad (9.5)$$

and the expression series satisfies

$$1 + V(z) = \exp \left(\sum_{j \geq 1} \frac{J(z^j)}{j} \right). \quad (9.6)$$

The term $-z^3$ removes $\sqrt{1}$. At every fixed degree all sums are finite. The same integer Euler-transform recurrence computes the coefficients. Every minimum-size expression avoids all these forbidden atoms, so V is still a valid coefficientwise upper model for minimum-size value counts.

9.3 What is now proved about the listed terms

Let L_n denote the cumulative number of certified sample values and let U_n^* be the cumulative upper count from (9.6). Then $L_n \leq a(n) \leq U_n^*$. Equality holds between these bounds for every $n \leq 21$. The remaining certified intervals are displayed below.

The unresolved gaps in this table do not weaken the exponential lower bound: the digit certificate needs only distinct values with known expression costs. Nor does the existence proof depend on trusting any particular OEIS term.

n	Certified lower bound L_n	Certified upper bound U_n^*	Gap
16	1,379	1,379	0
17	2,424	2,424	0
18	4,277	4,277	0
19	7,588	7,588	0
20	13,513	13,513	0
21	24,162	24,162	0
22	43,336	43,337	1
23	77,978	77,982	4
24	140,683	140,691	8
25	254,487	254,504	17
26	461,409	461,439	30
27	838,433	838,487	54
28	1,526,536	1,526,635	99

Table 2: Exact finite certificates. All terms from $n = 1$ through $n = 21$ are proved exactly by matching bounds. Completeness for $22 \leq n \leq 28$ is not claimed. The full table is included as `data/term_bounds.csv`.

10 Exact equality is decidable with an explicit bound

For completeness, we give a finite, entirely explicit alternative to an exact algebraic-number package. It is much too pessimistic for practical large computations, but establishes effectivity without numerical assumptions.

Proposition 10.1 (Uniform separation). *If $x \neq y$ are represented by expressions of size at most n , then*

$$|x - y| \geq (n + 1)^{-(2^{2n-2}-1)} =: \delta_n. \quad (10.1)$$

Proof. Adjoin all radical subexpression values from both expressions. The resulting field K has degree at most 2^{2n-2} : each expression has at most $n - 1$ radical nodes, each requiring an extension of degree at most two. The nonzero number $x - y$ is an algebraic integer, so $|\mathrm{N}_{K/\mathbb{Q}}(x - y)|$ is a positive integer, at least one. By Lemma 3.1, under every embedding of K its absolute value is at most the sum of the two leaf counts, hence at most $n + 1$. One factor in the norm is the specified real value $x - y$. Bounding all the other factors gives $1 \leq |x - y|(n + 1)^{[K:\mathbb{Q}]-1}$, which implies (10.1). $\square$

At precision P , fixed-denominator interval propagation as in (7.1) produces widths at most $2n2^{-P}$. One can see this inductively: addition adds widths, and square root on $[1, \infty)$ reduces a width by at least a factor of two before adding less than two units of outward rounding. Choose P so that

$$2^P > \frac{8n}{\delta_n}. \quad (10.2)$$

Every interval then has width less than $\delta_n/4$. If two such intervals overlap, their true values differ by less than $\delta_n/2$, so Proposition 10.1 forces equality. Disjoint intervals certify inequality. This gives a finite exact comparison algorithm for all expressions of bounded size.

Alternatively, real algebraic numbers represented by defining polynomials and isolating intervals provide exact equality and ordering; SageMath’s **AA** is a documented implementation [5]. The

present archive does not require SageMath. Its executed verifiers use only the Python standard library, and the lower-bound verifier does not need the astronomical worst-case precision (10.2) because it never declares overlapping intervals equal.

11 Interpretation and remaining asymptotic questions

The distinction between four levels of information is important. First, exponential upper and lower estimates only show that $\log a(n) = \Theta(n)$. Second, the injective-spine argument proves the stronger assertion that the coefficient of n in $\log a(n)$ converges. Third, the large-expression decomposition proves computability of that coefficient and the exact critical identity (2.5). Fourth, a full asymptotic equivalent would require further information that is not supplied by these arguments.

In particular, the following issues remain unresolved in this article:

- (i) A substantially tighter numerical enclosure of γ . The finite-data procedure is convergent, but its elementary upper codes are not optimized for speed, and complete exact enumeration is expensive.
- (ii) The existence of a ratio limit $a(n+1)/a(n) \rightarrow \gamma$. A root limit and monotonicity alone do not imply this.
- (iii) An asymptotic equivalent, including a power of n and a leading constant. The familiar $n^{-3/2}$ behavior of many tree models motivates a question, not a conclusion, for the quotient by all numerical identities.

It would be incorrect to identify the growth constant of an unordered-tree majorant with γ . The majorants deliberately keep numerically redundant expressions, and even rare local identities can change an exponential growth constant. Likewise, a fit to the first 28 numbers cannot certify a limiting base or a polynomial prefactor.

The rigorous outcome is

$$\log a(n) = n \log \gamma + o(n), \quad 1.86204 < \gamma < 1.9208, \quad \sum_{n \geq 1} a(n) \gamma^{-n} = \gamma^2.$$

The constant is characterized by convergent algebraic brackets computed from finite exact data. The finite numerical work in the archive checks the stated bounds independently of approximate equality tests.

A Reproducing the computations

The archive is self-contained apart from a Python 3.10 or later installation and a LaTeX installation for rebuilding the PDF. No computer-algebra package, internet connection, or floating-point equality oracle is needed for the executed certificates.

From the archive's top-level directory, run:

```
python verify_certificate.py --output data/verification_results.json
python certify_upper_bound.py --output data/upper_bound_certificate.json
python analyze_counts.py
```

The first command replays all expression nodes, checks symbol costs, and verifies pairwise disjoint enclosures. The second recomputes the formal upper coefficients, the strict convergence inequality, and the improved finite-term upper bounds. The third reproduces the rational lower-root brackets and the CSV table.

To regenerate the lower-bound sample from scratch:

```
python certified_enumerate.py --max-n 28 --precision 256 --output data
```

This generator is deliberately conservative about overlap. Its mathematical specification is a certified sample, not a complete exact enumeration. The independently replayable certificate is the primary finite witness.

In the execution used for this article, the sample generator completed in about five seconds and the independent sample verifier in about two and a half seconds. These are descriptive timings for that environment, not portable performance guarantees. The certificate itself has 1,526,536 records, nine bytes per record, plus a 16-byte header.

Files in the archive

File	Purpose
article.tex, article.pdf	This article and its source.
README.md, Makefile	Reproduction instructions and PDF build target.
certified_enumerate.py	Conservative interval-based construction of the sample.
verify_certificate.py	Independent integer-arithmetic replay and distinctness check.
certify_upper_bound.py	Exact coefficient recurrences and outward-rounded upper certificate.
analyze_counts.py	Rational root brackets and the finite-term bounds table.
data/digit_sample.bin	Compressed-by-sharing expression DAG; formula costs are charged in full.
data/certificate_metadata.json	Generator counts and certificate interpretation.
data/verification_results.json	Independently verified counts, minimum gap, and lower inequality.
data/upper_bound_certificate.json	Exact signed convergence witness and finite upper counts.
data/finite_lower_roots.json	Rational brackets for Table 1.
data/term_bounds.csv	All 28 lower and upper finite-term bounds.

B Certificate format and verification boundaries

The binary file begins with the eight ASCII bytes A158415D, followed by two little-endian unsigned 32-bit integers: interval precision and maximum assigned symbol cost. Each subsequent record consists of one unsigned byte for cost and two unsigned 32-bit child indices. The value 0xffffffff is a sentinel. The first record has cost one and two sentinels, representing 1. A later record whose right index is the sentinel represents the square root of its left child; otherwise it represents the sum of its two children. Child indices must precede the current record. The verifier recomputes each cost and enclosure rather than trusting them.

These checks establish concrete finite claims. The growth-limit, critical-identity, and computability theorems are justified by the mathematical proofs in the article; the archive does not contain a

proof-assistant formalization of them. Conversely, the numerical bounds do not depend on assuming that the theoretical seed constructions have been explicitly materialized as short expressions.

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